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Competitive AI Maturity Benchmarking (2025): Telecom Italia S.p.A. vs Peers

Outside-in Competitive Assessment

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Introduction and Context....4
High-Level Insights....4
Outcome Sought....5
AI Maturity KPI Framework (AlixPartners Model)....6
Competitive AI Maturity Benchmarking (2025): Telecom Italia S.p.A. vs Peers....8
Scoring Methodology....8
Category Analysis....10
Strategy....11
Execution....12
Technical....12
Organizational....13
Innovation....14
Overall Position Summary....14
Functional Insights & Best Practices by KPI Category....15
1. Strategy....15
Best Practices....15
Target Company Gaps....15
Improvement Levers....16
2. Execution....16
Best Practices....16
Target Company Gaps....16
Improvement Levers....17
3. Technical (Data & Infrastructure)....17
Best Practices....18
Target Company Gaps....18
Improvement Levers....18
4. Organizational (People, Structure, Governance)....19
Best Practices....19
Target Company Gaps....19

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Improvement Levers 19
5. Innovation 20
Best Practices 20
Target Company Gaps 20
Improvement Levers 20
SWOT 22
Strengths 22
Weaknesses 22
Opportunities 23
Threats 23
Cross-Dependencies 24
Sources 25
DISCLAIMER 32

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Introduction and Context

AI Maturity Benchmarking Report – Introduction and Context

Telecom Italia S.p.A. (TIM) operates in a European telecom market where margins are tight, network economics are shifting toward software, and regulation is becoming more prescriptive. Against that backdrop, TIM has moved beyond proofs of concept and is deploying AI across network operations, customer care, cybersecurity and digital channels, under a Board-approved set of Artificial Intelligence Guidelines and a multi-year technology plan that prioritizes data centers, edge capabilities and strategic partnerships. The Group has also concentrated capability in an AI hub in Turin and embedded cross-functional oversight to manage risk, ethics and compliance as AI scales. This positions TIM to translate platform investments into operational value while maintaining regulatory alignment. [2][12]

Peers are setting a fast cadence. Vodafone is industrializing AI on a unified platform and deepening hyperscaler alliances (Microsoft and Google), with measurable improvements in customer experience. Orange combines enterprise-scale deployment with a formal AI ethics council and is commercializing GenAI solutions for B2B clients. CK Hutchison is experimenting with agentic AI in frontline operations, while iliad is committing multi-billion-euro capital to AI infrastructure and launching AI-bundled propositions. Across Europe, the EU AI Act, the rise of AI-RAN for 5G-Advanced from 2026, growing demand for sovereign/regulated cloud, and a $500bn+ enterprise digitization opportunity are making AI central to competitiveness and growth in telecom. As of October 31, 2025, these market and technology currents define the context for TIM's next phase of AI maturity. [127][149][128][130][150][133][151][134][39][72][36]

High-Level Insights

TIM's strategic foundations for AI are credible and maturing. The company has codified Responsible-AI principles in a Board-approved guideline, set up centralized cross-functional oversight, and aligned its technology roadmap to data-center expansion, edge use cases and major platform partnerships. That scaffolding is translating into live deployments in the network, care and security domains and into selective customer-facing innovations. The governance spine is comparatively strong for the sector, yet publicly available materials do not cite a dedicated Chief AI Officer or a permanent Board-level AI/technology committee—leaving an execution gap between policy and single-point accountability that peers increasingly address through formalized leadership structures. In the near term, sustained advantage will depend on keeping ethics-by-design credible while elevating AI to a first-class business capability with named owners and enterprise KPIs. [2][57][50][12]

On value capture, TIM shows tangible impact but uneven instrumentation. Sparkle's AISNA platform reports materially faster operations

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(e.g., 30% reduction in NOC alert-handling time and significant automation of network updates), a Google-powered voice agent has been reported to improve care efficiency, and TIM S.A. in Brazil cites meaningful gains in customer-satisfaction scores from AI-enabled voice assistants. TIM also moved AI into the consumer proposition by bundling one year of Perplexity Pro access for customers in Italy, signaling appetite to differentiate on experience. Still, only a handful of use cases are disclosed with quantified outcomes, and enterprise-wide ROI measurement, causal methods, and portfolio-level reporting are not yet visible—constraining the ability to reallocate capital toward the highest-return AI initiatives. [41][32][14][55]

Technically, TIM's hard infrastructure is an advantage: a national data-center footprint with significant installed capacity, expansion at the edge, and multi-cloud elasticity through deep collaboration with Google (including Italian regions and a Turin innovation center). For public-sector and regulated workloads, TIM is advancing sovereign-cloud models that keep keys in country, aligning with European data-residency expectations. The primary constraints sit above the hardware: public disclosures emphasize risk and ethics controls but do not yet evidence industrialized, business-visible MLOps (model registry, CI/CD, drift/data-quality monitoring) or a unified, governed data operating model with a canonical schema and pervasive metadata/lineage. These gaps slow reuse and raise run-risk as the model estate grows. [69][67][72][50][52]

Relative to peers, the competitive bar is rising. Vodafone's platformization and multi-market scale produce a steadier drumbeat of productized AI, including GenAI-augmented assistants with documented NPS uplift; Orange pairs ethics leadership with repeatable B2B offers and internal GenAI adoption at scale; iliad's investment program aims to leapfrog on compute and sovereign stack; CK Hutchison is moving quickly in Asia with cloud-native and agentic pilots. For TIM, defensible differentiation by 2030 will come from converting today's governance and infrastructure into a repeatable value engine: codifying enterprise metrics and ROI, accelerating closed-loop network automation as 5G-Advanced/AI-RAN matures, and using sovereign-cloud and local-language assets to win regulated B2B and public-sector demand. The operators that combine transparent, audit-ready controls with high-velocity, measurable AI throughput will set the standard in European telecom. [127][128][152][150][151][133][39][72]

Outcome Sought

This benchmarking will deliver an evidence-based view of TIM's AI maturity—strategy, execution, technology, organization and innovation—relative to Vodafone Group, iliad, CK Hutchison and Orange. It will pinpoint where TIM is advantaged and where scale, operating discipline, or capability depth are limiting value realization; translate peer signals into concrete moves; and provide a prioritized 12–18-month roadmap that links use-case scale-up, data and MLOps industrialization, governance enhancements and KPI/ROI

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instrumentation to P&L outcomes. The intent is to help TIM accelerate efficiency, monetize differentiated experiences and B2B offers, and build the operating system—leadership, platform and controls—to capture AI-driven growth through 2030.

AI Maturity KPI Framework (AlixPartners Model)

To objectively benchmark AI maturity, we apply the AlixPartners five-category AI Maturity KPI Driver Framework, tailored to telecommunications. The model was designed to separate signal from noise in AI programs by combining verifiable operational evidence with qualitative calibration and peer normalization. For Telecom Italia S.p.A. (TIM), this means testing whether AI is institutionally embedded across the telco value chain—from network planning and operations to customer care, field services, enterprise products, and security—rather than evidenced only by pilots or marketing claims [2][55]. The framework is purpose-built to enable objective, comparable evaluation of AI readiness and performance: each category represents a critical pillar of enterprise AI, with driver-level KPIs that are reproducible, transparent, and open to external validation (e.g., board-approved policies, independently attested security controls, quantified case studies, patent filings) [41][50][142]. Because AI maturity evolves over time, the model explicitly balances strategic intent (e.g., published roadmaps), operational evidence (e.g., closed-loop automation in the network), and technical depth (e.g., MLOps, metadata, sovereign cloud) to track the shift from experimentation to measurable enterprise value. The telco adaptation emphasizes metrics that matter for operators: NPS/churn and ARPU uplift in customer channels, AHT/first-contact-resolution and deflection in care, MTTR/availability and energy-per-bit in the network, opex and capex productivity in field and infrastructure, and security/privacy resilience across data flows [143][144][131]. A reflective note is warranted: maturity cannot be inferred from surface-level counts of “AI use cases” or PoCs. In a regulated, mission-critical industry, what distinguishes leaders is institutionalization—clear decision rights, measurable outcomes tied to funding, lifecycle governance aligned to the EU AI Act, and platforms that turn models into repeatable releases at scale [50][129][142].

Strategy – what it measures and why it matters. Strategy tests whether AI is anchored to the corporate growth and margin thesis with explicit executive ownership, a time-bound roadmap, and disciplined prioritization. In telecoms, this includes the degree to which AI is positioned as a lever in network automation, care transformation, and new enterprise offerings (e.g., AI-enabled security, IoT, and edge services) [2][143]. Representative indicators include the existence and quality of a published AI roadmap; board-level oversight and director expertise; C-level advocacy tied to funding envelopes; clarity of use-case selection and business-case logic; timing of adoption; localization across markets (languages, regulation, data); and the presence of disruptive AI-native propositions beyond efficiency (e.g., agent assistants bundled with tariffs) [55][145]. High maturity in telco looks like a board-approved AI strategy with quantified targets (NPS, churn, energy, EBIT), a horizon-balanced portfolio, and market-specific playbooks for sovereign AI and compliance. Peers with scaled programs pair formal ethics governance with customer-facing AI that measurably drives CX and margin, while laggards show intermittent communications and follower

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timing [145][142].

Execution – the industrialization of value. Execution assesses the ability to build, integrate, and scale AI with measured returns. Scope spans funding sufficiency, build-buy-partner balance, the number and share of deployed and ROI-positive use cases, release cadence, business-visible monitoring (drift/data quality and business KPIs), closed-loop automation, portfolio discipline, and compliance-by-design (e.g., DPIAs, model cards) [41][143]. In telecoms, high performers industrialize AI like software: monthly release trains; stage-gated, finance-co-owned business cases; portfolio kill rates; enterprise model observability; and SLA-bound alert-to-action runbooks for live operations (e.g., self-healing network slices, dynamic care routing) [131]. Measurable signals include AHT/deflection improvements in care, MTTR/availability in the network, and energy savings per bit; vendor case studies and external audits provide triangulation but are weighted by credibility and recency [41][58]. The strategic implication is stark: without standardized value instrumentation and portfolio hygiene, even well-funded AI programs revert to islands of excellence rather than enterprise transformation [135][143].

Technical – the data and platform backbone. The Technical category covers enterprise data strategy and architecture, integration and interoperability, infrastructure scalability and performance (including edge), analytics/AI platforms, MLOps and model management, security/privacy, legacy modernization, and adoption of frontier tech (5G, O-RAN/RIC, edge, sovereign cloud). In telecoms, a unified, model-agnostic fabric—data catalog and glossary, canonical domain models, feature store, model registry, policy-as-code, automated lineage, CI/CD for ML, and real-time telemetry—unlocks reuse and speed, while sovereign cloud and Tier IV/edge capacity ensure resilience and regulatory fit [23][144][131]. High maturity features elastic hybrid/edge infrastructure with autoscaling, governed self-service marketplaces, and SLO-driven observability that ties model health to business KPIs; privacy/security are evidenced by ISO-aligned controls, DPO/DPIA regimes, and minimized incident exposure [50][146]. Leaders also exploit frontier tech—RIC/agentic controls in the network, AR/VR field support, and AI-for-energy optimization—on standardized platforms; laggards show fragmented stacks, limited metadata, and manual ML workflows that slow scale and increase risk [65][131].

Organizational – leadership, governance, talent, and culture. Organizational maturity examines whether structures and behaviors can sustain AI at scale. Indicators include the presence of a Chief AI (or Data & AI) officer or an equivalent federated model with named domain leads; board-approved Responsible-AI policies and ethics bodies; cross-functional governance and risk appetite by domain; role-based curricula with certification and tracked coverage; AI talent density and hiring; research team scale; employee well-being and change-management programs; international and multicultural collaboration; and transparency/assurance (e.g., dashboards, audits) [50][129][142]. In telecoms, high maturity embeds ethics-by-design (aligned to EU AI Act) into the SDLC, publishes KPIs, and ties incentives to adoption and risk outcomes. It also couples large-scale upskilling with visible leadership ownership and change infrastructure, recognizing union, regulator, and safety dynamics for critical networks [83][142]. Firms with strong governance but thin leadership benches or low AI

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talent density often struggle to close the strategy-execution gap; those with vibrant learning cultures and codified decision rights convert platforms into adoption faster [129][143].

Innovation – research, IP, and ecosystem shaping. Innovation measures how the company creates and diffuses new AI capabilities. Scope includes patenting and publications; open-source presence and an OSPO; joint labs and academic partnerships; structured experimentation (funnel KPIs, stage gates, reuse); standards and regulatory engagement; privacy-preserving data-sharing mechanisms; and vendor/partner rigor. In telecoms, defensible advantage accrues from domain-specific IP (e.g., radio optimization, fraud prevention), joint labs and sovereign AI initiatives, and open communities around RIC/xApps or data schemas; standardized, privacy-preserving data exchange (e.g., ISACs) accelerates learning without undue risk [57][81][147]. High maturity pairs a selective IP strategy with credible open-source contributions, measurable PoC-to-production conversion, and leadership roles in standards bodies; firms that rely solely on alliances or marketing narratives without IP, publications, or OS contributions risk imitability and vendor lock-in [55][145][142]. From strategy to execution, the pattern is consistent: robust innovation engines feed differentiated offerings and platform reuse, while weak engines confine AI to incremental efficiency gains.

Taken together, the five categories provide a multi-lens, telecom-specific view of AI maturity that is evidence-weighted, comparable across peers (Vodafone Group, iliad, CK Hutchison Group Telecom, Orange), and sensitive to regulatory context and sovereign-cloud realities. As the market shifts toward agentic operations and EU-AI-Act enforcement, the companies that institutionalize these five pillars—not just announce them—will convert AI from “projects” into compounding enterprise value [129][148][142].

Competitive AI Maturity Benchmarking (2025): Telecom Italia S.p.A. vs Peers

This benchmarking assesses Telecom Italia S.p.A. (TIM) against Vodafone Group plc, iliad SA, CK Hutchison Group Telecom Holdings Limited (CKHGT), and Orange S.A. over an evidence window spanning January 2024 through October 31, 2025. The objective is not simply to rank who is ahead; it is to surface the structural, operating, and cultural factors that explain performance differences and to translate those insights into practical priorities for leadership. The analysis uses the five-category AlixPartners AI Maturity Framework—Strategy, Execution, Technical, Organizational, and Innovation—scoring each on a 1–5 scale in which 1–2.4 = Low/Low-Medium, 2.5–3.4 = Medium, 3.5–4.2 = Medium-High, and 4.3–5.0 = High. Scores are triangulated from public programs, governance artifacts, platform signals, operating outcomes, and independent case evidence.

This matters now because telcos are at a pivotal convergence: network automation is becoming closed-loop, generative AI (GenAI) is re-wiring customer and employee interactions, and 5G/edge analytics is shifting compute to the network’s edge. Operators that marry clear

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strategy, scaled execution, industrialized data/ML platforms, and a learning organization will translate AI into lower opex/energy, faster time-to-care, better NPS, and new service lines; those that do not will see rising cost to compete, regulatory drag, and value leakage to hyperscalers and digital natives [153][152][136]. The EU AI Act and regional sovereignty demands reinforce the need for disciplined governance and localization, turning ethics-by-design and data residency into sources of speed rather than sources of delay [130][142].

Scoring Methodology

Scores are assigned on a 1–5 scale at the driver level and aggregated to category averages, with 1 = Low (ad-hoc or absent), 2 = Low-Medium (foundations in place, inconsistent), 3 = Medium (repeatable practices, partial institutionalization), 4 = Medium-High (scaled, governed, measured), and 5 = High (fully institutionalized, continuously optimized, externally validated). A 5.0 represents a state where AI is embedded into every relevant function, governed as a core capability, and evidenced by third-party assurance and sustained, quantified outcomes. Most telecom peers fall between 2 and 4 due to partial institutionalization across data governance, lifecycle automation, and portfolio value discipline [143][144].

Basis for evaluation. Each driver is scored from verifiable public evidence and corroborated qualitative signals: company filings and sustainability reports; investor and earnings materials; vendor case studies; standards/consortia participation; patent and publication databases; and press and policy disclosures. Examples include TIM's board-approved AI Guidelines and governance disclosures, Tier-IV/edge infrastructure and sovereign-cloud posture, quantified network and care case studies, and partner labs; comparable artifacts are collected for peers such as Vodafone and Orange to enable calibration [2][41][55][145][142]. Claims are weighted by time (recency) and source credibility; when vendor case studies are used, we adjust for marketing bias and seek triangulation (e.g., cross-market replication, independent audits). Contradictory items are resolved by preferring higher-confidence, more recent, and more primary sources.

Aggregation and calibration. Driver-level scores roll up to category averages (unweighted mean by default). Where a category comprises heterogeneous drivers with differing evidentiary strength, we apply qualitative calibration within a narrow band to avoid overstating thin signals or understating robust ones; any such adjustments are noted in the detailed driver commentary. Peer normalization is performed against a reference set (Vodafone Group, iliad, CK Hutchison Group Telecom, Orange) and a global "best practice" bar anchored in operators with platform-level MLOps, enterprise metadata/MDM, quantified ROI governance, and ethics-by-design embedded into the SDLC [135][129][143][142].

Temporal framing and scope. Assessments reflect the 2024–2025 evidence window with a cut-off of October 31, 2025; earlier artifacts are used for tenure-related drivers (e.g., first production year) and longitudinal impact where relevant. Because AI maturity is dynamic, scores

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should be interpreted as a point-in-time view rather than a static label.

Interpretation. The scoring is evidence-weighted, not perception-based. Quantitative drivers (e.g., number of deployed use cases, ROI-positive share, patents, publications) are balanced with qualitative drivers (e.g., cultural maturity, leadership intent), producing a nuanced measure of institutionalization. The purpose of benchmarking is not to "rank for rank's sake," but to surface actionable improvement levers and capability gaps—where strengthening strategy-to-execution linkage, codifying data foundations and lifecycle automation, and scaling responsible-AI governance will most accelerate value capture for TIM in the next 12–24 months [2][41][143][142].

Category Analysis

Category	Telecom Italia S.p.A.	Orange S.A.	Vodafone Group plc	CK Hutchison Group Telecom Holdings Limited	iliad SA	Gap Explanation
Strategy	3.2	3.7	3.8	3.3	3	TIM's strategy is coherent and governance-led—Board-approved AI Guidelines, a multifunctional oversight team, and a published technology plan and roadmap—yet board-level AI expertise, enterprise KPIs tied to funding, and market localization are incomplete. Vodafone and Orange score higher due to clearer enterprise platforms, multi-market impact narratives, and frequent C-level articulation; iliad's bold €3bn AI investment is newer and less institutionalized; CKHGT shows breadth but uneven institutionalization [2][3][8][62][65][154][153][142][151][155].
Execution	2.7	3.9	3.7	2.9	2.5	TIM integrates AI into several core processes and shows tangible efficiency wins (e.g., NOC and care), strong compliance-by-design, and a pragmatic release cadence. Gaps exist in enterprise ROI discipline, business-visible model monitoring, external benchmarking, and portfolio stage-gating. Orange and Vodafone

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						outpace peers on industrial scale and cadence; CKHGT and iliad are scaling but with thinner value instrumentation [41][12][15][69][66][68][143][135][158][134].
Technical	2.8	3.7	3.6	3.2	2.9	TIM's hard infrastructure and privacy/security are strong (Tier IV/edge footprint, multi-cloud, privacy-by-design). The operating layer—enterprise data leadership, canonical data model, metadata/MDM, governed self-service, and automated MLOps—remains partial. Orange and Vodafone show more industrialized platforms across markets; CKHGT is solid on infra/security but fragmented on platforms; iliad's compute lead outpaces its MLOps/governance maturity [23][26][21][73][75][25][72][160][161][162][163].
Organizational	3.4	4.2	3.9	3.3	2.7	TIM's governance spine is strong—Board-approved AI Guidelines, integrated ERM, and ethics-by-design—backed by significant learning programs and change mechanisms. Gaps are structural: no CAIO, limited named AI leadership bench and disclosed headcount, and limited memberships in AI-ethics bodies. Orange leads with an Ethics Council and scaled enablement; Vodafone is close behind with enterprise adoption and cultural readiness; CKHGT is solid but talent and research scale trail; iliad is earlier in governance institutionalization [42][46][44][55][34][37][38][45][142][165][166].
Innovation	3.2	3.7	3.1	2.2	2.9	TIM's IP engine and alliances are credible—High on AI patenting and Medium-High on academic engagement and vendor rigor—but open-source contributions are minimal and experimentation is not yet industrialized. Orange combines heavy patenting and deep academic ties; Vodafone excels at ecosystem scaling but has a thinner proprietary IP footprint; iliad's infrastructure and open-science signals are strong but early; CKHGT is partnership-led with limited upstream IP/open activity [57][56][83][61][79][168][169][170][171].

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Strategy

The clarity and credibility of the AI ambition: alignment to corporate goals, leadership ownership, funding, governance, and market/segment localization.

Analysis

TIM sits in the Medium tier (3.2/5). Strengths include a formal AI policy framework approved by the Board, a cross-functional governance model, and visible investments (e.g., Turin AI hub, sovereign-cloud posture, and consumer-facing GenAI via the Perplexity Pro offer) that bind AI to experience and margin aims [2][3][13]. Live deployments span network operations and customer care, signalling that strategy is not paper-only [12]. Where it trails the leaders is the scaffolding around the ambition: few directors with AI credentials, no dedicated AI board committee, uneven localization beyond EU compliance, and a lack of enterprise AI KPIs tied to capital allocation and workforce scaling [1][10][65]. By contrast, Vodafone's decade-long platformization and recently expanded hyperscaler alliances provide a clear architecture for market-level scaling and impact storytelling, even if board-level RAI oversight is still evolving [149][127][154][156]. Orange combines strong ethics governance (Data & AI Ethics Council), a pragmatic AI narrative across network, CX, and enterprise, and regional localization via sovereign platforms, albeit without a fully time-bound, public roadmap [130][153][157]. CKHGT demonstrates portfolio breadth and new ventures but with more recent proof of durability; iliad has announced outsized AI investments and launched AI-native propositions, yet remains early in institutional governance and measured outcomes [155][151][134]. For TIM, the strategic implication is to close the governance-to-value loop: add AI-literate board oversight, publish a 24–36-month roadmap with milestones and owners, standardize AI KPIs linked to funding, and expand localization beyond regulatory alignment to multilingual models and market-specific datasets [2][3][62][65].

Execution

The ability to operationalize AI at scale: integrate into workflows, ship at cadence, monitor/model-manage in production, and measure ROI at portfolio level.

Analysis

TIM's Execution score (2.7/5) reflects solid enablement and pockets of impact offset by operating-system gaps. Evidence points to embedded AI in customer service, network monitoring, and cyber, with quantifiable gains—Sparkle's AISNA reduced NOC alert-handling time by ~30% and cut execution time of massive updates by up to 80%; Google-powered care agents report ~20% efficiency improvement [41][32]. Compliance is institutionalized (DPIA, DSAR tooling, multi-jurisdiction coverage), supporting safe rollout at scale [69]. However, only

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a handful of use cases are documented with hard financial outcomes; portfolio counts are low relative to narrative breadth; business-visible model health (drift, data quality, incident SLOs) is nascent; release velocity is episodic; and there is no standardized, auditable ROI methodology with finance sign-off [13][67][17][68]. Orange leads on breadth, cadence, and proportion ROI-positive, underpinned by an enterprise platform and cross-functional embedding—though it too can strengthen causal ROI methods and external delivery benchmarks [143][152]. Vodafone demonstrates industrialized MLOps and frequent functional releases across markets, but value instrumentation and external benchmarking are not yet pervasive [135][159]. CKHGT shows momentum (e.g., agentic AI in Hong Kong care, group-wide rollouts) but likewise needs enterprise-grade ROI and monitoring [158][155]. Iliad's releases and B2C GenAI activation are visible, yet enterprise embedding and measurement are still maturing [134]. For TIM, four moves would convert enablement into repeatable value: institutionalize ex-ante/ex-post ROI with a central benefits ledger; deploy business-visible model observability with SLA-bound alert-to-action runbooks; introduce external delivery benchmarks for speed/quality/cost; and stage-gate the portfolio with horizon mix and kill criteria [12][66][68].

Technical

Data and compute foundations, analytics/AI platforms, MLOps rigor, interoperability, security/privacy, and emerging-tech leverage.

Analysis

TIM's Technical profile is split between a robust chassis and a maturing operating system. On the plus side, the group operates large-scale, AI-ready infrastructure—17 data centers with Tier IV sites, sovereign/public cloud integration, an edge program for low-latency workloads—and maintains certified privacy/security with privacy-by-design [23][26]. Data and AI are present across multiple domains (CX, network, cyber), and emerging tech adoption (IoT, edge, Open RAN) is active [21][28]. Gaps are systemic: no enterprise CDO mandate beyond the DPO, no enterprise-wide canonical data model, thin active metadata/lineage and MDM, limited governed self-service, and only partial CI/CD, registry, and automated monitoring across models [19][73][75][72][25]. By comparison, Orange and Vodafone operate more standardized AI platforms and pipelines at multi-market scale, even as each still works through canonical data and metadata gaps [160][161]. CKHGT shows an AI-ready, secure hybrid core with improving integration but fragmented analytics platforms and uneven MLOps [162][164]. Iliad has built top-tier compute and sovereign cloud capacity but must now institutionalize data/ML ops to translate infra into repeatable throughput [163]. For TIM, the unlock is organizational and architectural: appoint a Chief Data & AI with a federated data council; define a canonical model and business glossary; deploy an enterprise catalog/marketplace with policy-as-code and lineage; standardize model registry/CI-CD/observability; and finish targeted legacy integration to remove interoperability bottlenecks [21][23][73].

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Organizational

Talent depth, leadership accountability, ethics and risk governance, AI literacy, culture and change, and international collaboration.

Analysis

With a 3.4 score, TIM's organization is more mature than its technical and execution cores. The company embeds ethics-by-design into policy and governance, with Board-approved AI Guidelines, centralized oversight, and integration into ERM; whistleblowing, audit, and corrective-action loops reinforce continuous improvement [42][44][55]. Learning programs are large-scale (e.g., 280k hours, AI Academy), and change management and well-being programs are active [34]. Where TIM under-signals is ownership and depth: no Chief AI Officer, few publicly named AI leaders, limited validated AI headcount and external ethics-body participation [37][38][45]. Orange's independent Ethics Council, Responsible-AI design authority, and enterprise-scale enablement push it into the upper Medium-High tier; Vodafone's cross-functional operating model and literacy give it similar altitude, even as both can deepen role-based certification and quantified risk appetite [142][165]. CKHGT's policies, change frameworks, and collaboration are well-formed, but talent density and leadership breadth are modest; iliad's culture is mobilized, yet governance and role clarity are still consolidating [166][167]. For TIM, the next step is to convert governance into scale capacity: appoint a CAIO or federated domain leads; publish a model inventory and AI-specific KPIs; tie role-based certification to progression; and seek external assurance or ISO/IEC 42001 to harden trust signals [42][46].

Innovation

Research output, patents, open innovation and alliances, standardized data-sharing, standards/regulatory engagement, and open-source posture.

Analysis

TIM scores Medium (3.2). A WIPO scan attributes ~58 AI-related patent families over the past five years—High by our rubric—and the company maintains a broad academic and vendor ecosystem (e.g., joint labs, EU projects, Turin innovation center) [57][56]. Alliances and vendor-assessment rigor are structured and risk-based [83]. Offsetting this, open-source presence is limited, peer-reviewed publication volume is modest relative to ecosystem breadth, and the experimentation funnel lacks portfolio-level metrics and stage-gates [61][79]. Orange sits Medium-High on the back of strong patenting, academic links, privacy-preserving data-sharing and alliances (e.g., sovereign GenAI platforms), though it also has scope to scale open source and industrialize experiments [168][172]. Vodafone's innovation model is "build-with and scale": ecosystem execution and internal platformization are strong while proprietary IP and open-science signals are lighter

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[169][135]. Iliad's €3bn AI push and the Kyutai open-science lab give it a compelling upstream story that still needs scaled outputs; CKHGT remains partnership-heavy with limited IP or open-source leadership [170][151][171]. For TIM, the route to a higher tier is to create an OSPO with a small number of flagship, maintained AI repos; set a publication strategy; measure PoC-to-production hit rates; and instrument alliances with co-investment and outcome KPIs [79][83].

Overall Position Summary

Overall, TIM's AI maturity averages ~3.06/5 (15.3/25), ranking third of five peers: Orange leads at ~3.84 (19.2/25), Vodafone follows at ~3.62 (18.1/25), then TIM (~3.06), CKHGT (~2.98), and iliad (~2.80). The pattern is consistent across categories: TIM's Organizational capability (3.4) is its relative strength; Strategy and Innovation are competitive (3.2 each) but trail Orange and Vodafone; Technical (2.8) and Execution (2.7) are the binding constraints. In concrete terms, TIM possesses a strong governance spine—Board-approved AI Guidelines, ERM-linked ethics-by-design, scaled learning—and a credible patent/alliances engine [42][46][55][57]. Yet platform industrialization and value discipline lag: no enterprise CDO/CAIO, limited canonical data/metadata, nascent model observability, few ROI-audited use cases, and incomplete localization beyond EU compliance [19][73][67][68][65]. Industry dynamics make these gaps consequential. Telecom AI economics now depend on closed-loop network automation, GenAI-enabled care/sales, and edge analytics, all of which require unified data/ML ops, portfolio hygiene, and board-level risk ownership to satisfy the EU AI Act and to contain energy and compute costs [153][136][130]. Strategically, this benchmarking suggests TIM is "governance-ready but operating-constrained." To compete against Orange's ethics-anchored enterprise engine and Vodafone's platformized scaling, TIM should: 1) elevate data/AI leadership and standardize MLOps; 2) harden value capture with finance-co-owned ROI and portfolio gates; 3) modernize data foundations (canonical model, lineage, catalog) and governed self-service; 4) deepen localization with multilingual and market-specific models; and 5) open its innovation aperture via an OSPO and a metricized experimentation funnel [21][23][68][61]. Done in 2025–2027, these moves would shift TIM from Medium toward Medium-High and build the platform needed for AI-driven cost, quality, and growth advantages through 2030.

The next section, Category Deep Dives, unpacks the evidence and root causes behind each score and sets out the focused, no-regrets actions to move Telecom Italia from governance-ready to platform-industrialized—and from pockets of impact to enterprise-scale value—over the 2025–2027 horizon.

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Functional Insights & Best Practices by KPI Category

1. Strategy

Definition: Measures how well AI ambition, leadership ownership, and governance are aligned with enterprise goals.

Best Practices

Target Company Gaps

Telecom Italia (TIM) has a coherent, governance-led AI agenda with board-approved AI Guidelines and a published Technology Plan, but gaps remain where leaders excel. First, oversight depth: no director has proven AI credentials and there is no dedicated board AI/Technology committee, limiting strategic challenge capacity and signaling risk oversight remains generalist [1][3]. Second, outcome transparency: while use-case impact is evidenced in pockets, enterprise-wide AI KPIs tied to funding and business-unit scorecards are not disclosed; public messaging is periodic, not programmatic [12][5][62]. Third, market fit: localization focuses on EU compliance rather than multilingual models and segment-specific datasets, capping differentiation beyond the home market [65]. Fourth, portfolio breadth and timing: only four documented use cases with earliest operational AI in 2020 imply follower adoption and limited diversification relative to peers [6][7]. Strategically, these gaps risk slower scale-up, governance opacity for investors and regulators, and missed growth optionality from AI-native products.

Quick Evidence Callouts:

- Board-approved AI Guidelines formalize ethics and governance [2][3].
- Three-year Technology Plan sets objectives and oversight cadence [12].
- No AI-expert director and no board AI committee evident [1].
- Localization: strong on EU compliance, thin beyond EU [65].
- Four disclosed use cases; follower adoption circa 2020 [6][7].

Improvement Levers

1. **Institutionalize AI governance at board level and cascade to scorecards -** Create a Board AI & Technology Committee chaired by an AI-literate non-executive and mandate quarterly AI risk–return reviews across markets. Expand director competencies via targeted recruitment and a briefed advisory network to raise the quality of challenge. Cascade committee decisions into BU scorecards by tying

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investment envelopes to a small set of enterprise AI KPIs (NPS, churn, cost-to-serve, energy per bit, revenue uplift) with CEO/CFO ownership. Use the existing Responsible-AI Guidelines as the policy backbone and add committee-level thresholds for high-risk systems (model cards, DPIAs, red-teaming) [3]; mirror precedents where ethics councils and pan-regional AI policies anchor strategy [130][129]. This lever shifts AI from principles to auditable, investor-grade accountability and will compress cycle times on strategic choices. Impact: High Example: Stand up the Board AI & Technology Committee in 90 days; publish its remit and the initial KPI set with Q1/Q3 progress updates.

2. **Publish a 24-month AI roadmap and KPI ledger, with market-level localization -** Consolidate an enterprise AI roadmap that names the top 6–10 scaled use cases by P&L, the investment envelope per case, and quarterly milestones; link each to a KPI ledger owned by Finance. Require ex-ante baselines and ex-post impact reporting and publish an external synopsis annually alongside sustainability disclosures [12]. Add a localization plan that implements multilingual models, regional datasets, and fairness checks in priority markets, moving from EU-compliance-only to segment-specific fit [65]. Use marquee customer propositions (e.g., assistant bundling) to prove monetization hypotheses in consumer segments while expanding B2B AI offers [1]. The effect is a visible, time-bound strategy that reduces uncertainty, sustains budget continuity, and signals discipline to capital markets. Impact: High Example: Within 180 days, release the roadmap, KPI ledger, and a localization playbook; report progress in quarterly calls.

2. Execution

Definition: Focuses on operational scaling, feedback loops, and value capture across live AI use cases.

Best Practices

Target Company Gaps

TIM has crossed from pilots to production with solid governance and selected efficiency wins, yet the value engine is inconsistent. Only a few use cases are documented, and even fewer carry quantified, finance-verified outcomes; there is no standardized, auditable ROI methodology [13][14][68]. External delivery benchmarks are absent, and business-visible model monitoring appears limited to technical telemetry rather than executive dashboards with SLA-to-action workflows [66][67]. Portfolio discipline is nascent: stage-gates and kill criteria are not evidenced, and agentic architectures are early [70][10]. The consequence is value leakage — funding continues without systematic proof or managed reprioritization — which explains the gap to peers with programmatic cadence and enterprise telemetry.

Quick Evidence Callouts:

- Four documented AI implementations since 2023 [13].

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- Two quantified impact cases: AISNA efficiency and voice agent AHT [14].
- No enterprise, auditable ROI methodology disclosed [68].
- Limited external benchmarking for delivery metrics [66].
- Business-visible model health not evidenced at scale [67].

Improvement Levers

3. **Stand up a finance-anchored benefits-realization system -** Co-own AI economics with Finance. Require ex-ante baselining (A/B or diff-in-diff), counterfactual design, and cost capture for every initiative; enforce ex-post verification and variance reviews; and publish a central benefits ledger rolled into BU OKRs. Tie gate approvals and funding tranches to KPI attainment and codify a quarterly portfolio review. Use the two quantified exemplars as templates and normalize them across network, care, sales, and IT workflows [14]. This lever converts enablement into demonstrable P&L outcomes and creates the evidence base for capital allocation and investor narratives. Impact: High Example: Launch the ledger and standardized business-case template in 60 days; link Q2 budgets to gate criteria.
4. **Make model health business-visible and action-oriented -** Deploy a unified observability layer spanning data quality, performance, drift, fairness, and business KPIs; expose role-based dashboards to product owners and executives; and connect alerts to SLA-backed runbooks. Register all models with owners and risk tiers; require model cards and DPIAs at release; and add red-team tests for high-risk systems. This closes the loop between technical signals and business action, reduces incidents, and shortens time-to-fix, mirroring closed-loop practices seen in network automation at peers [136]. The result is fewer outages, faster recovery, and higher stakeholder confidence. Impact: High Example: Within 120 days, inventory 100% of production models; pilot dashboards in care and network ops with weekly exec reviews.

3. Technical (Data & Infrastructure)

Definition: Emphasizes infrastructure, data integration, and lifecycle industrialization that enable scalable AI.

Best Practices

Target Company Gaps

TIM's hard infrastructure and privacy/security controls are solid — Tier-IV DCs, edge readiness, and a mature privacy program underpin scaled deployments — but the data and ML operating system is under-powered. There is no enterprise CDO mandate; metadata and canonical data structures are immature; self-service via a governed catalog is limited; and CI/CD, registries, and automated model monitoring are not evidenced at scale [19][73][72][25][71]. Inorganic accelerants are sparse (few AI investments/acquisitions), risking slower

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capability build and higher vendor dependence [29][30]. The strategic risk is scale friction — duplicated effort, inconsistent model quality, higher run costs, and avoidable compliance exposure — which constrains time-to-value versus peers with unified foundations.

Quick Evidence Callouts:

- No enterprise-wide CDO role evidenced [19].
- Metadata/canonical model maturity remains Low-Medium [73][75].
- MLOps tooling/processes are not automated at scale [25][71].
- Governed self-service portal limited in scope [72].
- Security/privacy strong but largely policy-not-code [26].

Improvement Levers

5. **Appoint a Chief Data & AI Officer and stand up a federated data operating model -** Name a CDAIO with authority over data strategy, stewardship, and AI lifecycle standards; launch a Data & AI Council with BU domain owners; and publish a canonical data model, business glossary, and ownership map. Mandate lineage and quality SLOs for critical flows and integrate privacy controls as code at ingestion. Tie program funding to adoption of standards; use sovereign and hyperscaler footprints as distribution, not fragmentation, points [131]. This lever creates a single spine for interoperability and accelerates reuse across network, care, and enterprise offers. Impact: High Example: In 120 days, formalize the CDAIO remit; release v1 canonical entities (customer, product, network event) and a glossary with 80% BU coverage.
6. **Standardize the ML lifecycle and observability as platform capabilities -** Converge on a reference MLOps stack (registry, feature store, CI/CD, experiment tracking, canary deploys) and make it the default path for all models. Instrument performance, drift, cost, and fairness telemetry; automate rollback and retraining triggers; and require model cards and DPIAs at release. Package these as shared platform services and measure adoption. The goal is to move from policy-heavy to automation-heavy control, mirroring the platform industrialization seen at leaders [135][132]. This will cut time-to-production, reduce incidents, and improve run-rate economics. Impact: High Example: Within 180 days, onboard the top 10 models to the new stack; publish SLOs and monthly adoption/incident metrics.

4. Organizational (People, Structure, Governance)

Definition: Assesses human and structural enablers: leadership, talent, governance, culture, and change management for AI at scale.

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Best Practices

Target Company Gaps

TIM's governance spine, ethics, change and learning programs are advanced, yet structural ownership and talent density lag. There is no CAIO and few publicly named AI leaders; verified AI headcount is small as a share of the workforce, and external affiliation with AI-ethics bodies is absent [37][36][38][45]. AI-specific enterprise KPIs, model inventories, and business-visible risk dashboards are not published despite strong oversight structures [42]. The risk is execution drag: governance outpaces delivery capacity and external credibility without visible ownership, depth, and measured outcomes. Compared with peers operating independent ethics councils and scaled enablement, this keeps TIM just below a High tier.

Quick Evidence Callouts:

- Board-approved AI Guidelines; cross-functional oversight [42].
- No CAIO; limited named AI leadership [37][36].
- Low validated AI headcount share; hiring pipeline growing [38][33].
- No memberships in AI-ethics bodies disclosed [45].
- Large-scale training hours; AI Academy in place [34].

Improvement Levers

7. **Create a CAIO-led, federated AI leadership model with shared KPIs -** Appoint a Chief AI Officer reporting to the CEO and chairing a Group AI Council of domain leads (Network, Care, Commercial, Enterprise, Security). Define a model inventory and a compact set of AI KPIs (CX, churn, opex/energy, revenue, risk) that cascade into BU scorecards; review them quarterly alongside risk dashboards. Integrate RAI gates (bias, privacy, robustness) into the SDLC with clear decision rights. This clarifies accountability, aligns incentives, and turns governance into measurable performance — echoing peers that have institutionalized independent oversight and cross-functional control [130]. Impact: High Example: Within 100 days, appoint the CAIO and publish the Group AI Council charter and KPI pack; begin quarterly reviews in Q2.
8. **Scale role-based certification and visible external assurance -** Evolve the Academies into role-based certification paths (execs, product, data, engineering) and set annual completion targets for priority roles; link certification to progression. Seek external assurance (e.g., ISO/IEC 42001 readiness assessment) and join at least one recognized AI-ethics consortium to strengthen stakeholder trust [34]. Publish participation and skills-coverage metrics and include well-being indicators to ensure a just transition. This elevates talent density, signals

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seriousness to partners and regulators, and accelerates adoption quality. Impact: Medium Example: Publish role-based curricula and 12-month coverage targets in the next sustainability filing; begin an ISO/IEC 42001 gap assessment in H1.

5. Innovation

Definition: Explores forward-looking R&D, ecosystem collaboration, IP generation, open science, and standardization that sustain long-term advantage.

Best Practices

Target Company Gaps

TIM's innovation profile mixes real invention with underpowered openness and transfer. On the strength side, the company has a high AI-patent count and a broad alliance portfolio [57][80]. Gaps concentrate in open-science levers: minimal open-source presence, modest peer-reviewed publication volume, and standards engagement that is participatory rather than shaping [61][60][82]. Experimentation is structured but not yet instrumented with funnel KPIs to raise hit rates and reuse [79]. Strategically, this limits developer mindshare, slows knowledge diffusion, and reduces influence over emerging norms — areas where peers are beginning to lean in. Notably, TIM's overall Innovation score is slightly above the peer average, but with clear upside in openness and transfer.

Quick Evidence Callouts:

- ~58 AI-related patent families in 5 years [57].
- Broad research partnerships and joint labs [56].
- Low open-source contribution footprint [61].
- 13 AI papers (2023–2025) — modest volume [60].
- Structured but mid-maturity PoC-to-product process [79].

Improvement Levers

9. **Launch an Open-Source Program Office (OSPO) with 3–5 flagship AI releases -** Stand up an OSPO to govern contributions, licensing, and security; select 3–5 non-core assets (e.g., data schemas, evaluation harnesses, domain adapters) and release them with roadmaps, maintainers, and contribution guidelines. Tie OSS to recruitment and ecosystem strategy, measure stars/forks and downstream citations, and ensure legal alignment with Responsible-AI policies. This raises developer mindshare, accelerates learning, and positions TIM in

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communities that shape the stack — complementing a strong patent base [57]. Impact: Medium Example: Within 180 days, publish the OSPO charter and the first two repositories with maintainer SLAs and quarterly adoption metrics.

10. **Industrialize the pilot-to-product funnel and amplify scientific presence -** Define stage-gates (problem fit, data readiness, security/RAI, unit economics) and enforce go/kill criteria; track hit rate, time-to-production, and value realized. Pair this with a publication strategy: target 6–8 peer-reviewed papers per year from live pilots and standards pilots; publish model cards and reference implementations to convert learning into influence [79][82]. Embed tech-transfer playbooks in alliances to secure shared IP and co-marketing, turning lab wins into product growth. The result is faster diffusion, stronger credibility, and clearer ROI from R&D spend. Impact: High Example: Publish funnel KPIs and targeted venues for H2; require a “reference implementation” package for each successful pilot within 60 days of go-live.

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SWOT

Strengths

• Board-approved AI governance with centralized oversight and ethics-by-design. TIM's Artificial Intelligence Guidelines are formally approved at Board level, align with the EU AI Act/GDPR/UNESCO principles, and define roles across development, procurement, and operations. This governance spine reduces regulatory risk and speeds approvals by clarifying decision rights and embedding compliance upfront, creating a trusted foundation for scaling AI across consumer, enterprise, and network domains. [50][57][46] ([50], [57], [46])

• AI embedded in core processes with early closed-loop automation and quantified wins. Sparkle's AISNA shows 30% faster NOC alert handling and up to 80% faster network updates, while TIM's Google-powered voice agent cut call-handling effort by ~20%. Participation in the AI@EDGE program signals closed-loop automation readiness, connecting detection to autonomous actions. These proofs anchor the shift from pilots to production and create credible opex and CX baselines for further scale. [41][32][22] ([41], [32], [22])

• Elastic, sovereign-ready hybrid infrastructure and multi-cloud partnerships. A national data-centre footprint (including Tier IV sites), expanding edge cloud and sovereign-control models, and deep ties with Google Cloud provide latency, capacity and sovereignty options for AI workloads. This mix de-risks cost/performance while enabling rapid deployment of network and customer AI at national scale. [67][69][50] ([67], [69], [50])

• Scaled workforce enablement through an AI Academy and large-scale reskilling. TIM delivered 280,000 training hours to 14,400 employees in 2024 and operates branded academies (including AI) with certified paths. Building internal fluency reduces vendor dependence and accelerates adoption by frontline teams, improving change velocity and benefit realization. [50][90][83] ([50], [90], [83])

Weaknesses

• No Chief AI Officer and limited AI expertise at Board level. Without a CAIO or a Board technology/AI committee, accountability is diffused and strategy-to-execution translation slows. This gap raises execution risk on prioritization, funding discipline and cross-unit alignment for scaled AI. [37][4] ([37], [4])

• Sparse enterprise-wide outcome instrumentation and low documented use-case depth. Only four implementations are publicly documented and just two cite quantified impact, with no standardized, auditable ROI/KPI framework. This constrains resource reallocation and undermines credibility versus peers with portfolio-level value tracking. [13][14][68] ([13], [14], [68])

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• Immature enterprise MLOps and limited business-visible model health and metadata foundations. Evidence of automated model registries, CI/CD, drift monitoring and role-based dashboards is thin, and canonical data/metadata maturity is low-medium. The result is slower reuse, higher run risk and harder AI Act compliance at scale. [25][73][75] ([25], [73], [75])

• Localization leans on EU compliance, not multilingual/multi-market optimization; adoption timing is follower. Governance maps well to EU rules, but there is little proof of systematic multilingual data/model parity or market-specific tuning, and initial production adoption dates to 2020. That limits cross-market replication velocity and CX lift. [65][7] ([65], [7])

Opportunities

• Convert EU-AI-Act readiness into a commercial trust moat with measurable assurance. TIM can productize its governance—model cards, DPIAs, supplier clauses—into verifiable artifacts for customers and regulators, differentiating in sensitive verticals (public sector, health, finance). A trust-led go-to-market can raise win rates and reduce sales cycles. [69][46] ([69], [46]) [Impact: Mid-term]

• Monetize AI-enhanced consumer bundles and loyalty features at scale. The nationwide Perplexity Pro offer can evolve from promotion to tiered bundles, usage-based upsell and churn reduction levers integrated with TIM's CDP and retention models—turning AI engagement into ARPU defense. [55][21] ([55], [21]) [Impact: Mid-term]

• Industrialize network closed-loop automation for opex, energy and resilience gains. Extending AISNA-like assurance and AI@EDGE patterns across fixed/mobile domains can compress MTTR/energy per bit and stabilize quality during traffic shocks, freeing capacity and capital for growth plays. [41][22] ([41], [22]) [Impact: Long-term]

• Institutionalize an AI operating system (CAIO + ROI gates + standardized MLOps). A central owner, enterprise ROI methodology, and a common MLOps stack with stage-gates/kill criteria would lift portfolio hit-rate, cut cycle times and enable at-scale reuse across BUs—unlocking step-change productivity. [68][25][70] ([68], [25], [70]) [Impact: Mid-term]

Threats

• Vodafone and Orange are scaling AI faster across markets, compressing CX and cost advantages. Vodafone's 10-year Microsoft alliance and Orange's plug-and-play GenAI offers signal industrialized platforms and monetization motion; if uncountered, rivals set higher NPS/cost bars that shift share and investor narratives. [127][150] ([300], [816]) [Risk: High]

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• iliad's AI-bundled propositions intensify price-feature competition in core markets. Mass-market inclusion of an AI assistant (e.g., Le Chat Pro) pressures incumbents to match features and value, risking margin dilution if differentiation remains thin and cost bases are higher. [134][173] ([510], [470]) [Risk: Medium]

• Vendor concentration and thin inorganic AI investment elevate lock-in and talent risks. A partnership-heavy model without recent AI venture/M&A activity reduces control over roadmaps and IP while weakening talent magnetism, raising long-run unit-costs and strategic dependence. [11][29] ([11], [29]) [Risk: Medium]

• AI-Act enforcement and reputational risk intensify if observability lags deployment scale. Absent business-visible model health and standardized evidence artifacts, scaled rollouts raise exposure to fines and trust erosion when failures occur, slowing approvals and partner onboarding. [69][25] ([69], [25]) [Risk: High]

Cross-Dependencies

Governance strength can be the flywheel that unlocks scale: Board-approved AI Guidelines (S) shorten approval paths for network automation and consumer AI trials, partially offsetting the absence of a CAIO and fragmented role clarity (W). The elastic, sovereign-ready hybrid infrastructure (S) directly enables the network automation opportunity (O) while also raising the stakes on MLOps and observability gaps (W), which, if unaddressed, amplify compliance and reputational threats under the EU AI Act (T). Early, quantified wins in care and NOC (S) provide the proof points to institutionalize an AI operating system with ROI gates and standardized MLOps (O), which in turn mitigates vendor lock-in risk and the value-measurement deficit (W/T). Finally, consumer exposure via Perplexity (O) can blunt iliad's AI-bundled challenge (T) if TIM converts promotion into retention-accretive bundles backed by CDP-driven personalization and transparent impact KPIs—an execution loop that turns today's scattered successes into a repeatable advantage by 2030.

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